

The Binomial Random Walk

In the Bitcoin whitepaper, Satoshi Nakamoto models the vulnerability of the network to a double-spending attack as a stochastic process. Specifically, the race between an honest chain and an attacker's secret chain is characterized as a **Binomial Random Walk**.

The Variables

- p : Probability that an honest node finds the next block.
- q : Probability that the attacker finds the next block ($p + q = 1$).
- z : The number of blocks (confirmations) the honest chain has progressed.
- q_z : The probability that an attacker will ever catch up from z blocks behind.

The Gambler's Ruin Formula

The core of this analysis relies on the classic Gambler's Ruin problem. In this scenario, a gambler (the attacker) starts with a deficit and bets against an entity with infinite wealth (the honest network).

The probability q_z is the "hitting probability" the chance that the random walk will eventually hit the origin (zero gap) despite a disadvantageous drift. It is derived from the boundary conditions of a linear recurrence relation:

$$q_z = \begin{cases} 1 & \text{if } p \leq q \\ \left(\frac{q}{p}\right)^z & \text{if } p > q \end{cases}$$

This formula shows that if the attacker has less than 50% of the hashrate ($p > q$), their probability of success decreases exponentially as the lead z increases.

The Poisson Approximation

Because the attacker mines simultaneously with the honest network, the number of blocks the attacker finds by the time the honest nodes reach z blocks is not fixed. It follows a **Poisson Distribution** with an expected value λ :

$$\lambda = z \left(\frac{q}{p}\right)$$

Total Success Probability

To find the total probability of the attacker overtaking the chain, we sum the Poisson probability of the attacker finding k blocks multiplied by the Gambler's Ruin probability of catching up the remaining distance ($z - k$):

$$P = \sum_{\{k=0\}}^{\{\infty\}} \frac{\lambda^k e^{-\lambda}}{k!} \cdot \begin{cases} \left(\frac{q}{p}\right)^{z-k} & \text{if } k \leq z \\ 1 & \text{if } k > z \end{cases}$$

Security Implications

This model justifies why Bitcoin users wait for multiple confirmations. As z grows, the “Gambler” (attacker) is increasingly likely to be “Ruined” by the honest network’s majority power.

Confirmations (z)	Success Probability ($q = 0.3$)
0	1.0000
1	0.4162
2	0.2320
5	0.0573
10	0.0061

—

References

- Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*.
- Random walk - Wikipedia
- Binomial Random Walks and Bitcoin - Mathematics Stack Exchange